

University Important questions

Q1. Classify Triangles of the Neck

Q2. Anterior Triangle of the Neck

Q3. Submandibular Triangle

Q4. Occipital Triangle

Q5. Describe the boundaries and contents of a subclavian triangle

Q6. Carotid Sheath: Formation, Attachment, Relation, and Content

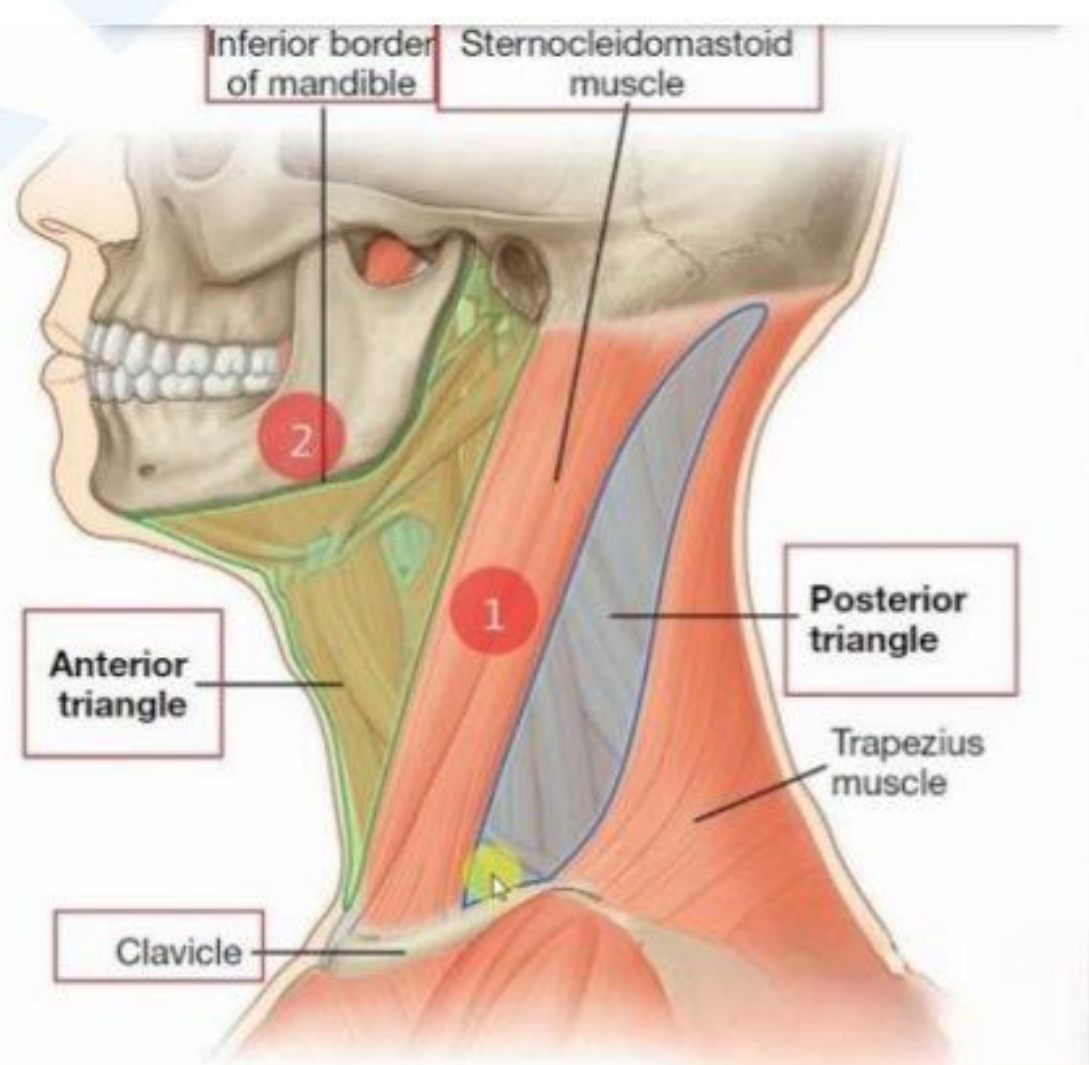
Q7. Sternocleidomastoid Muscle: Attachment, Nerve Supply ,Actions and relations of sternocleidomastoid

Q8. External Jugular Vein: Formation, Course, and Termination

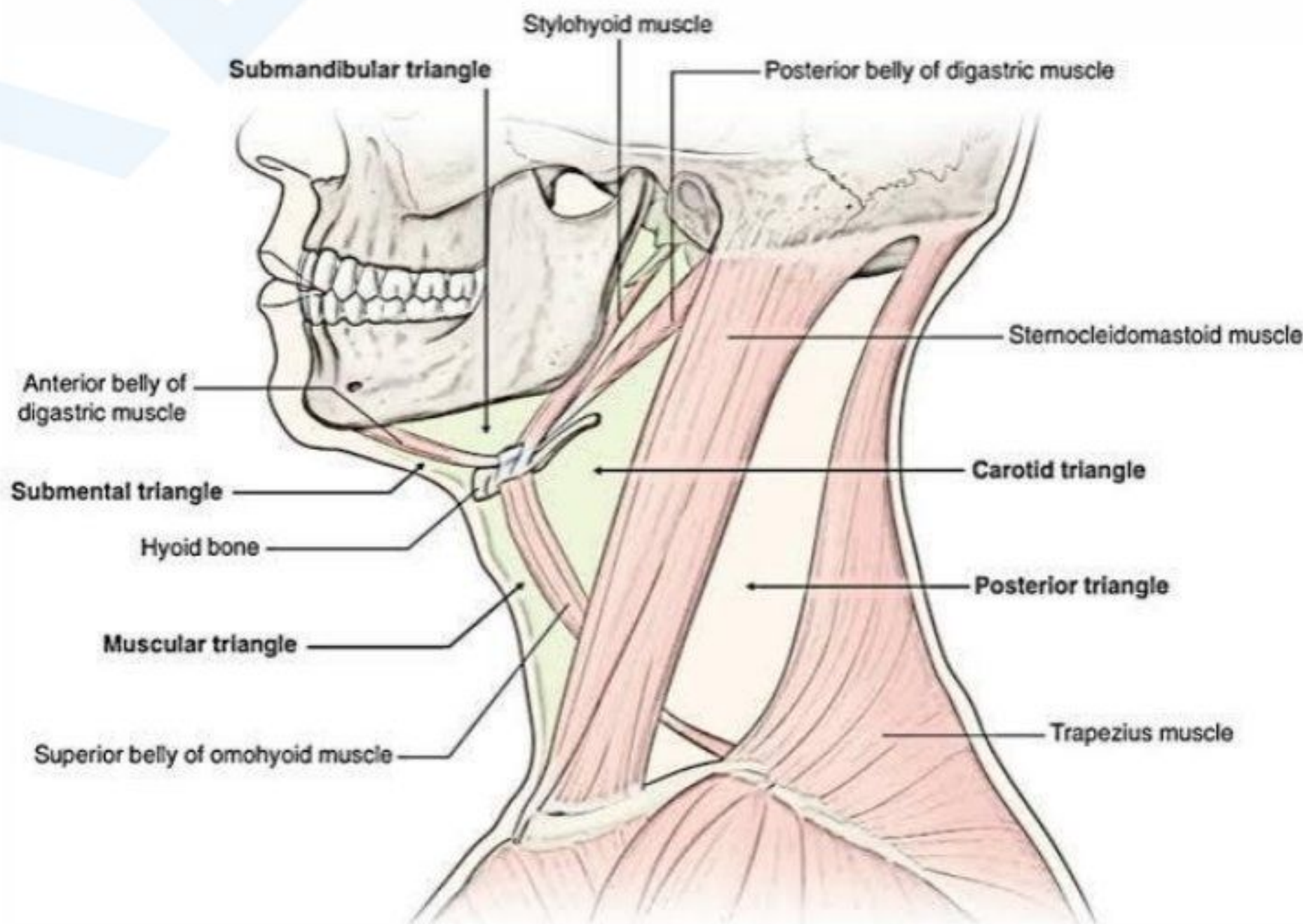
The most important questions are:

Q1, Q2, Q6, Q7

Q1. Classify Triangles of the Neck



Triangle	Boundaries	Subdivisions
Anterior Triangle	Anteriorly: Midline of the neck Posteriorly: Anterior border of the sternocleidomastoid muscle Superiorly: Lower border of the mandible	-Submandibular Triangle - Carotid Triangle - Muscular Triangle - Sublingual Triangle
Posterior Triangle	Anteriorly: Posterior border of the sternocleidomastoid muscle Posteriorly: Anterior border of the trapezius muscle Inferiorly: Middle third of the clavicle	- Occipital Triangle - Subclavian Triangle



Q2. Anterior Triangle of the Neck

Anterior Triangle of the Neck

Boundaries:

- **Anteriorly:**
 - Midline of the neck
- **Posteriorly:**
 - Anterior border of the **sternocleidomastoid muscle**
- **Superiorly:**
 - Lower border of the **mandible**

Subdivisions:

- **Submandibular Triangle/ Digastric Triangle (reason of its alternative name that is digastric triangle: Named because it is bordered by the two bellies of the digastric muscle.)**
 - **Boundaries:**
 - Anterior: Anterior belly of the **digastric muscle**
 - Posterior: Posterior belly of the **digastric muscle**
 - Inferior: Body of the **mandible**
 - **Contents:**
 - **Submandibular gland**
 - **Facial artery and vein**
 - **Lymph nodes**

- **Carotid Triangle:**
 - **Definition:** A triangular area in the neck that contains important **vascular** and **nervous structures**.
 - **Boundaries:**
 - Anterior: Superior belly of the **omohyoid muscle**
 - Posterior: Anterior border of the **sternocleidomastoid muscle**
 - Superior: Stylohyoid and digastric muscles
 - **Contents:**
 - **Common carotid artery** (which bifurcates into external and internal carotid arteries).
 - **Internal jugular vein.**
 - **Vagus nerve (CN X).**
 - **Accessory nerve (CN XI).**
 - **Hypoglossal nerve (CN XII).**
 - **Clinical Relevance:**
 - Important for **vascular surgeries**.
 - Palpation of the **carotid pulse** is done here.
 - Site of **carotid artery stenosis**.
- **Muscular Triangle:**
 - **Boundaries:**
 - Anterior: Midline of the neck
 - Posterior: Superior belly of the **omohyoid muscle**
 - Superior: Lower border of the **sternocleidomastoid muscle**
 - **Contents:**
 - **Thyroid gland**
 - **Parathyroid glands**
 - **Trachea and esophagus**
- **Sublingual Triangle:**
 - **Boundaries:**
 - Anterior: Body of the **mandible**
 - Posterior: Anterior belly of the **digastric muscle**
 - **Contents:**
 - **Sublingual gland**
 - **Lingual nerve**
 - **Hypoglossal nerve (CN XII)**

Q3. Submandibular Triangle

Submandibular Triangle

Location:

- Situated beneath the **mandible**.

Boundaries:

- **Anterior Boundary:**
 - Anterior belly of the **digastric muscle**
- **Posterior Boundary:**
 - Posterior belly of the **digastric muscle**
- **Superior Boundary:**
 - Lower border of the **mandible**

Contents:

- **Gland:**
 - **Submandibular gland:** Major salivary gland located within this triangle.
- **Nerves:**
 - **Hypoglossal nerve (CN XII):** Innervates the muscles of the tongue.
 - **Lingual nerve:** Branch of the mandibular nerve supplying the floor of the mouth and tongue.
- **Blood Vessels:**
 - **Facial artery:** Supplies blood to the face; gives off branches like the submental artery.
 - **Facial vein:** Drains blood from the face.
- **Lymph Nodes:**
 - **Submandibular lymph nodes:** Drain lymph from the oral cavity, teeth, and neighboring structures.

Clinical Significance:

- **Surgical Access:**
 - Important in procedures involving the submandibular gland and surrounding structures.
- **Infections:**
 - Awareness of this triangle is crucial in diagnosing infections such as **sialadenitis** (infection of the salivary gland) or **abscesses**.

Relations:

- **Superficial Structures:**
 - Covered by the **skin** and **platysma muscle**.
- **Deep Structures:**
 - Related to the **mylohyoid muscle**, which forms the floor of the mouth.

Q4. Occipital Triangle

Occipital Triangle

Boundaries:

- **Anterior Boundary:**
 - Posterior border of the **sternocleidomastoid muscle**
- **Posterior Boundary:**
 - Anterior border of the **trapezius muscle**
- **Inferior Boundary:**
 - Superior belly of the **omohyoid muscle**

Contents:

- **Nerves:**
 - **Accessory Nerve (CN XI):**
 - Innervates the **sternocleidomastoid** and **trapezius muscles**
 - **Cervical Plexus Branches:**
 - **Lesser Occipital Nerve:** Supplies sensation to the skin behind the ear
 - **Great Auricular Nerve:** Supplies sensation to the skin over the parotid gland
 - **Transverse Cervical Nerve:** Supplies sensation to the anterior neck
- **Vessels:**
 - **Occipital Artery:**
 - Branch of the **external carotid artery** supplying the posterior scalp
 - **Transverse Cervical Artery:**
 - Supplies muscles of the neck and shoulder
 - **External Jugular Vein:**
 - Drains blood from the posterior part of the head and neck
- **Muscles:**
 - **Splenius Capitis:**
 - Extends and rotates the head
 - **Levator Scapulae:**
 - Elevates the scapula and assists in neck extension
 - **Scalene Muscles:**
 - **Anterior, Middle, and Posterior:**
 - Aid in neck flexion and assist with respiration by elevating the ribs

Clinical Significance:

- **Nerve Blocks:**
 - Useful for performing **regional anesthesia** for neck surgeries and pain management
- **Surgical Access:**

- Important landmark in surgeries involving the neck and posterior head
- **Lymphatic Drainage:**
 - Involves lymph nodes that are critical in assessing infections or tumors in the head and neck region

Relations:

- **Superficial Structures:**
 - Lies deep to the **platysma muscle** and is covered by the **skin**
- **Deep Structures:**
 - Related to the **cervical plexus**, brachial plexus roots, and other major vessels in the neck

Q5. Describe the boundaries and contents of a subclavian triangle.

Subclavian Triangle

- **Small lower part of posterior triangle** of neck
- **Posterior triangle** → behind **sternocleidomastoid muscle**
- **Divided by inferior belly of omohyoid** into:
 - **Occipital triangle** (upper part)
 - **Subclavian (supraclavicular) triangle** (lower part)

Boundaries of Posterior Triangle

- **Anterior** → **posterior border of sternocleidomastoid**
- **Posterior** → **anterior border of trapezius**
- **Inferior (base)** → **middle one-third of clavicle**
- **Apex** → at **superior nuchal line** (where **trapezius & sternocleidomastoid meet**)

Roof

- **Formed by investing layer of deep cervical fascia**

Floor

- **Formed by prevertebral layer of deep cervical fascia**
- **Covers muscles:**
 - **Splenius capitis**
 - **Levator scapulae**
 - **Scalenus medius & sometimes scalenus posterior**

Contents of Subclavian Triangle

Nerves

- **Three trunks of brachial plexus** → emerge between **scalenus anterior & medius**
- **Nerve to serratus anterior (C5, C6, C7)** → arises by **3 roots**, passes over serratus anterior in axilla
- **Nerve to subclavius (C5, C6)** → descends in front of brachial plexus & subclavian vessels
 - **May give accessory phrenic nerve** → joins phrenic nerve in front of scalenus anterior
- **Suprascapular nerve (C5, C6)** → from upper trunk of brachial plexus
 - **Crosses posterior triangle**
 - **Reaches scapula** → supplies **supraspinatus & infraspinatus muscles**

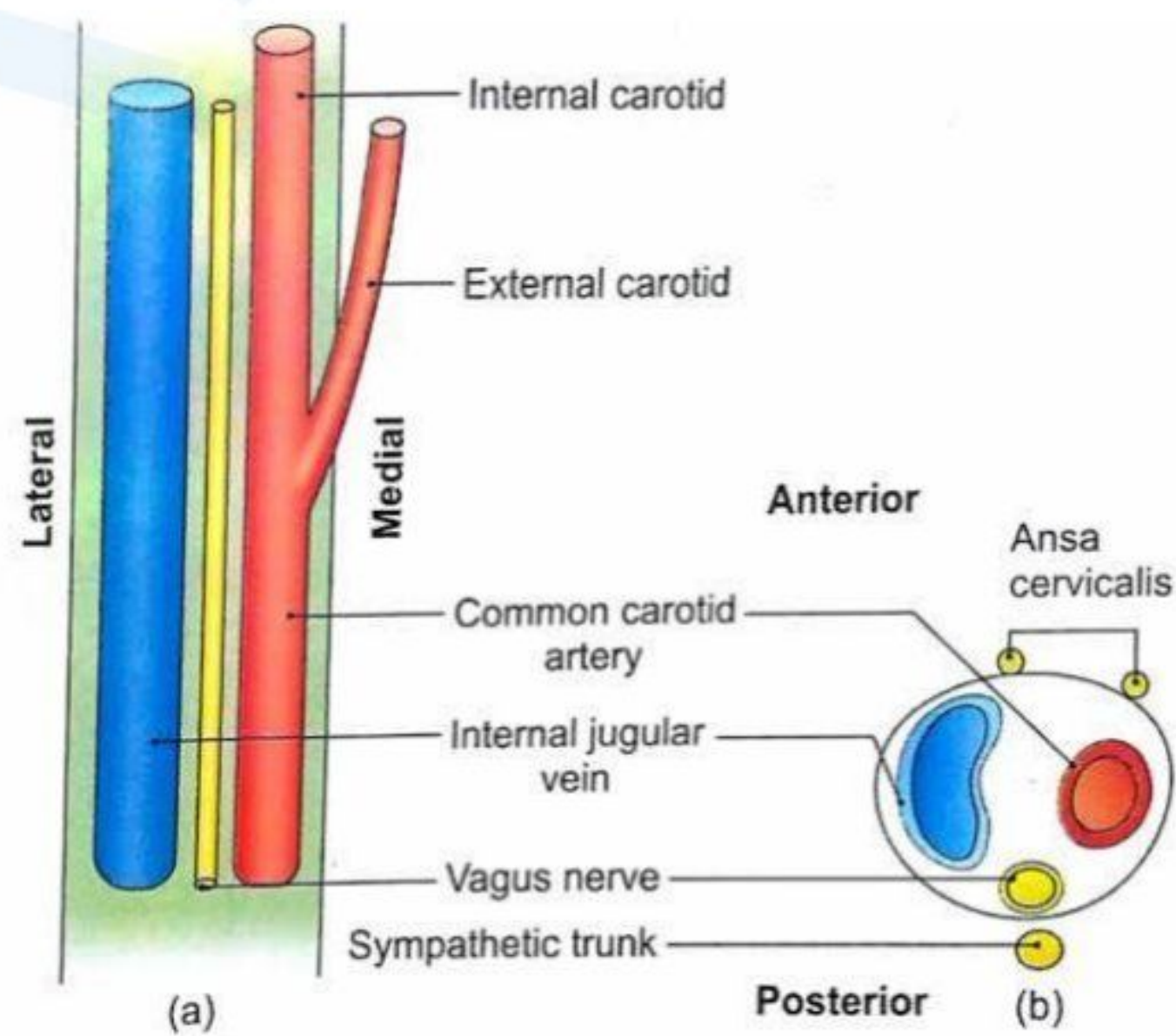
Vessels

- **Subclavian artery** → behind tendon of scalenus anterior
- **Subclavian vein** → in front of tendon
- **Suprascapular artery** → branch of thyrocervical trunk
 - **Passes laterally & behind clavicle**
- **Transverse cervical artery** → branch of thyrocervical trunk
 - **Crosses scalenus anterior, phrenic nerve, & upper trunks of brachial plexus**
 - **Divides into superficial & deep branches** at anterior border of levator scapulae

Lymph Nodes

- **Few supraclavicular lymph nodes**

Q6. Carotid Sheath: Formation, Attachment, Relation, and Content



Figs 3.8a and b: Right carotid sheath with its contents: (a) Surface view, and (b) sectional view

Formation:

- Formed by **deep cervical fascia**
- Composed of **investing layer** of cervical fascia

Attachments:

- Extends from the **base of the skull** to the **superior mediastinum**
- Encases important structures in the neck

Relations:

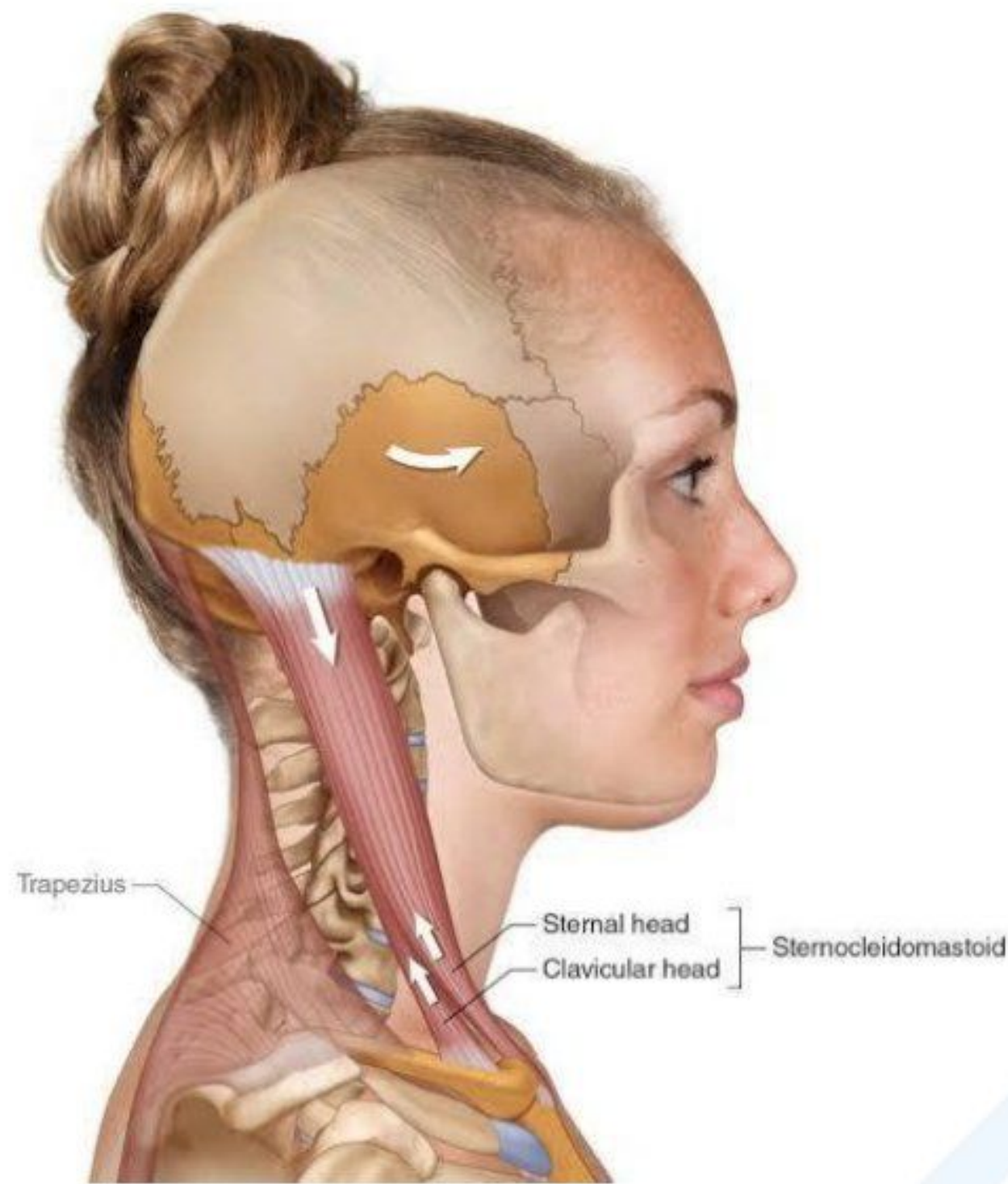
- **Anteriorly:**
 - Common carotid artery
 - Internal jugular vein
- **Laterally:**
 - Sternocleidomastoid muscle
- **Medially:**
 - Trachea and esophagus
- **Posteriorly:**
 - Prevertebral fascia and cervical spine

Content:

- **Common carotid artery** (and its bifurcation)
- **Internal jugular vein**

- **Vagus nerve (CN X)**
- **Sympathetic nerves** (cervical sympathetic ganglia)
- **Lymph nodes** (cervical lymph nodes)

Q7. Sternocleidomastoid Muscle: Attachment, Nerve Supply ,Actions and relations of sternocleidomastoid.



Attachment:

- **Origin:**
 - **Sternal Head:** Anterior surface of the **manubrium sterni**
 - **Clavicular Head:** Upper surface of the **medial third of the clavicle**
- **Insertion:**
 - Lateral surface of the **mastoid process** of the temporal bone
 - **Superior nuchal line** of the occipital bone

Nerve Supply:

- Innervated by the **accessory nerve (CN XI)**
- Additional sensory innervation from the **cervical plexus** (C2 and C3)

Actions:

- **Bilateral Contraction:**
 - Flexes the neck forward
 - Extends the head at the atlanto-occipital joint
- **Unilateral Contraction:**

- Rotates the head to the **opposite side**
- Tilts the head to the **same side**
- Assists in **elevating the thorax** during forced inhalation

Relations:

- **Anteriorly:**
 - Covers the **internal jugular vein** and **common carotid artery**
- **Posteriorly:**
 - Related to the **trapezius muscle**
- **Medially:**
 - In relation to the **phrenic nerve** and **brachial plexus**
- **Laterally:**
 - Lies close to the **external jugular vein**
- **Inferiorly:**
 - Overlies structures of the **anterior triangle of the neck**, including the **thyroid gland** and **trachea**

Q8. External Jugular Vein: Formation, Course, and Termination

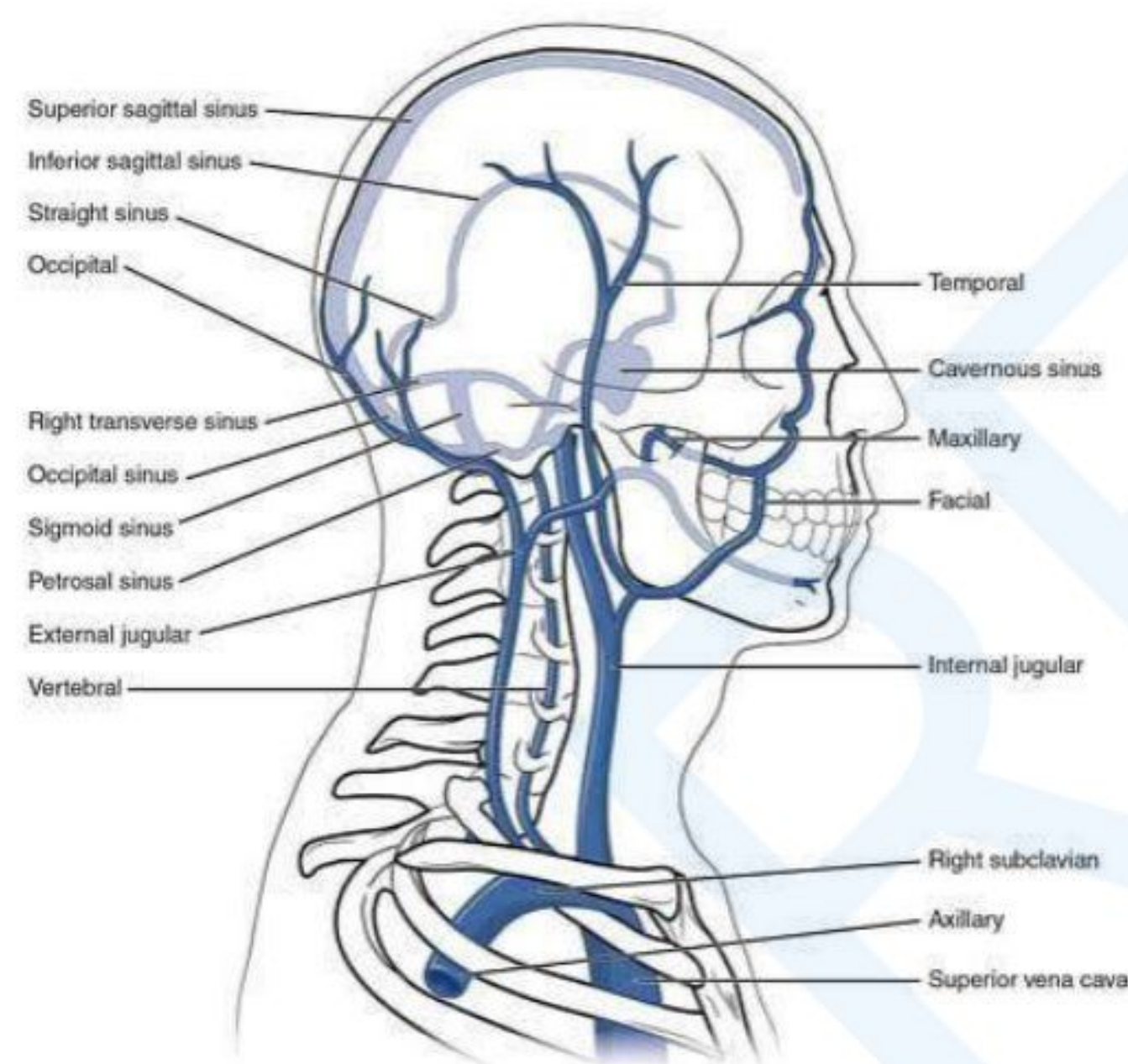


Fig. Veins of head and neck

Formation:

- Formed by the union of the **posterior auricular vein** and the **retromandibular vein**
- Emerges from the **posterior aspect** of the angle of the mandible

Course:

- Passes **downward** and **anteriorly** across the **sternocleidomastoid muscle**
- Lies **superficial** to the sternocleidomastoid muscle and deep to the **platysma muscle**
- Travels in a **vertical direction** along the side of the neck
- Receives tributaries from the **cervical and facial regions**

Termination:

- Drains into the **subclavian vein**
- Usually enters the subclavian vein just **above the clavicle**
- Sometimes may drain into the **internal jugular vein**